An explicit critical-density configuration for simultaneous Patterson–Sullivan reconstruction

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13 August 2026

Status: claimed full new solution, likely valid, pending human review. An explicit deterministic configuration is constructed. It answers the sharpened question (1.41) of Bufetov–Qiu and supplies an explicit instance for the deterministic-set problem in the original paper.

1 Source question and result

The source paper is A. I. Bufetov and Y. Qiu, *Patterson–Sullivan measures for point processes and the reconstruction of harmonic functions*, arXiv:1806.02306v2 [1]. Section 2.4, printed/PDF page 13, asks for explicit deterministic subsets of the disk with the simultaneous reconstruction properties established there for random point configurations.

2.4. **Open problems.** It is natural to suggest the following open problems.

- Construct in deterministic way explicit subsets $X \subset \mathbb{D}$ satisfying the simultaneous reconstruction properties as in Theorems 2.3, 2.5 or 2.8.
- Give geometric sufficient conditions for subsets $X \subset \mathbb{D}$ satisfying the simultaneous reconstruction properties as in Theorems 2.3, 2.5 or 2.8.
- Similar problems for complex hyperbolic spaces of higher dimensions.

Figure 1: The original open-problem list in arXiv:1806.02306v2, page 13.

The same authors sharpened the deterministic part in *The Patterson–Sullivan Interpolation of Pluriharmonic Functions for Determinantal Point Processes on Complex Hyperbolic Spaces*, arXiv:2101.09622v1 [2]. Their Question (1.41), printed page 7 (PDF page 17), asks whether there is an explicit configuration $X \in \mathrm{Conf}(\mathbb{D})$ of critical upper density such that

$$f(z) = \lim_{s \downarrow 1} \frac{g_X(s,z;f)}{g_X(s,z)} \quad \text{for every } f \in H^\infty(\mathbb{D}) \text{ and } z \in \mathbb{D},$$

where

$$g_X(s,z;f) = \sum_{x \in X} e^{-s d_{\mathbb{D}}(z,x)} f(x), \qquad g_X(s,z) = g_X(s,z;1).$$

We use the source normalization

$$d_{\mathbb{D}}(u,v) = \log \frac{1 + |\varphi_u(v)|}{1 - |\varphi_u(v)|}, \qquad \varphi_u(v) = \frac{v-u}{1-\overline{u}v}.$$

Let $P_z(\xi) = (1-|z|^2)/|\xi - z|^2$ be the Poisson kernel on $\mathbb{T}$, and let m be normalized Lebesgue measure on $\mathbb{T}$.

Question. *Can we construct in deterministic ways explicit configurations $X \in \mathrm{Conf}(\mathbb{D})$ with critical upper density (namely, the number of points of X inside any compact subset of $\mathbb{D}$ is controlled by the hyperbolic area of that compact subset) such that it satisfies the simultaneous interpolation property as follows:*

$$f(z) = \lim_{s \to 1^+} \frac{g_X(s, z; f)}{g_X(s, z)} \quad \text{for all } f \in H^\infty(\mathbb{D}) \text{ and all } z \in \mathbb{D}\,? \tag{1.41}$$

Here $H^\infty(\mathbb{D})$ is the space of all bounded holomorphic functions on $\mathbb{D}$.

Figure 2: The sharpened deterministic question (1.41) in arXiv:2101.09622v1, printed page 7.

Theorem 1 (Explicit dyadic-shell reconstruction)**.** *For $n \geq 1$, set*

$$M_n = 2^n, \qquad R_n = \log M_n, \qquad r_n = \tanh(R_n/2) = \frac{M_n - 1}{M_n + 1},$$

and define the completely explicit configuration

$$X_{\mathrm{dyad}} = \left\{ r_n e^{2\pi i j/M_n} : n \geq 1,\ 0 \leq j < M_n \right\}.$$

Then X_{dyad} is uniformly separated and has critical hyperbolic upper density. More precisely,

$$\#(X_{\mathrm{dyad}} \cap B_{\mathbb{D}}(z, R)) \leq C e^R \quad (z \in \mathbb{D},\ R \geq 1),$$

while $\#(X_{\mathrm{dyad}} \cap B_{\mathbb{D}}(0, R_n)) = 2^{n+1} - 2$.

For every compact $K \Subset \mathbb{D}$,

$$\lim_{s \downarrow 1} \sup_{z \in K} \sup_{\substack{f \text{ bounded harmonic on } \mathbb{D} \\ \|f\|_\infty \leq 1}} \left| \frac{g_{X_{\mathrm{dyad}}}(s, z; f)}{g_{X_{\mathrm{dyad}}}(s, z)} - f(z) \right| = 0. \tag{1}$$

In particular, the answer to Question (1.41) is affirmative.

2 Idea of the proof

At hyperbolic radius $R_n = n \log 2$ the construction places exactly 2^n equally spaced points. Thus its counting exponent is the critical exponent one. For $s > 1$, the contribution of the nth shell has the Abel weight $2^{-n(s-1)}$.

A bounded harmonic function is the Poisson integral of an $L^\infty(\mathbb{T})$ boundary function. The reconstruction statement therefore reduces to convergence of a positive boundary kernel. If the shell were continuously distributed in angle, its boundary kernel would be the Poisson convolution of P_z^s, and approximate-identity convergence would finish the argument. Replacing the circle by 2^n roots of unity introduces Fourier aliases near the nonzero multiples of 2^n. These aliases do not become small shell by shell: this is precisely the critical-density difficulty. Instead, aliases from different shells are almost orthogonal. An alias from shell n at the frequency scale 2^k, $k > n$, is suppressed by $r_n^{2^k} \leq \exp(-c 2^{k-n})$. The normalized Abel coefficients have squared sum $O(s-1)$, so the total alias error tends to zero in $L^2(\mathbb{T})$. This L^2, hence L^1, convergence closes the simultaneous quantifier over the entire unit ball of bounded harmonic functions.

3 Proof

3.1 Geometry and absolute convergence

Points on different shells are at distance at least $|R_n - R_k|$, so adjacent shells are separated by $\log 2$. On one shell, adjacent points satisfy

$$\sinh \frac{d_{\mathbb{D}}(r_n e^{2\pi i j/M_n}, r_n e^{2\pi i (j+1)/M_n})}{2} = \frac{2r_n \sin(\pi/M_n)}{1-r_n^2} = \frac{M_n^2-1}{2M_n}\sin\frac{\pi}{M_n} \geq \frac{3}{4}.$$

Thus X_{dyad} is uniformly separated. Disjoint hyperbolic balls of a fixed small radius around its points give, by the usual packing argument,

$$\#(X_{\text{dyad}} \cap B_{\mathbb{D}}(z,R)) \leq Ce^R.$$

The exact count at R_n is $\sum_{k=1}^n 2^k = 2^{n+1}-2$, proving criticality.

If f is bounded and z is fixed, the triangle inequality gives

$$\sum_{x \in X_{\text{dyad}} \cap \{|x|=r_n\}} e^{-sd_{\mathbb{D}}(z,x)}|f(x)| \leq \|f\|_\infty e^{sd_{\mathbb{D}}(0,z)}, 2^{-n(s-1)}.$$

Hence all sums in Theorem 1 converge absolutely for $s > 1$.

3.2 The shell weights

Write $\zeta_{n,j} = e^{2\pi i j/M_n}$ and put

$$a_{n,z}(\zeta) = e^{R_n - d_{\mathbb{D}}(z, r_n\zeta)}.$$

If $q = |\varphi_z(r\zeta)|$, then

$$e^{-d_{\mathbb{D}}(z,r\zeta)} = \frac{1-q}{1+q} = \frac{(1-|z|^2)(1-r^2)}{|1-\overline{z}r\zeta|^2(1+q)^2}.$$

Since $e^{R_n} = (1+r_n)/(1-r_n)$, it follows that

$$a_{n,z}(\zeta) = \frac{(1-|z|^2)(1+r_n)^2}{|1-\overline{z}r_n\zeta|^2(1+|\varphi_z(r_n\zeta)|)^2}. \tag{2}$$

Consequently, uniformly for z in a fixed compact subset of $\mathbb{D}$ and $\zeta \in \mathbb{T}$,

$$a_{n,z}(\zeta) = P_z(\zeta) + O_K(2^{-n}). \tag{3}$$

The same estimate holds after taking powers s uniformly for $1 \leq s \leq 3/2$.

Put $t = 2^{1-s}$ and $w_{s,z} = P_z^s$. Define the true normalized shell kernel

$$F_{n,s,z}(\eta) = \frac{1}{M_n}\sum_{j=0}^{M_n-1} a_{n,z}(\zeta_{n,j})^s P_{r_n\zeta_{n,j}}(\eta).$$

Then the unnormalized boundary kernel and its mass are

$$\mathcal{K}_{s,z} = \sum_{n\geq 1} t^n F_{n,s,z}, \qquad D_{s,z} = \int_{\mathbb{T}} \mathcal{K}_{s,z}\, dm = g_{X_{\text{dyad}}}(s,z). \tag{4}$$

We next isolate the only nonstandard estimate.

3.3 A dyadic quadrature lemma

Lemma 2 (Almost orthogonality of critical aliases)**.** *Let $M_n = 2^n$, $r_n = (M_n - 1)/(M_n + 1)$. Suppose a family $w_\lambda \in L^2(\mathbb{T})$ satisfies, uniformly in λ,*

$$|\widehat{w}_\lambda(k)| \le A\rho^{|k|} \quad (k \in \mathbb{Z})$$

for some $A < \infty$ and $0 < \rho < 1$. Set

$$Q_{n,\lambda}(\eta) = \frac{1}{M_n}\sum_{j=0}^{M_n-1} w_\lambda(\zeta_{n,j}) P_{r_n\zeta_{n,j}}(\eta),$$

$$C_{n,\lambda}(\eta) = \int_{\mathbb{T}} w_\lambda(\zeta) P_{r_n\zeta}(\eta)\, dm(\zeta), \qquad E_{n,\lambda} = Q_{n,\lambda} - C_{n,\lambda}.$$

There are constants $C, c > 0$, independent of n, k, λ, such that

$$\|E_{n,\lambda}\|_2 \le C, \qquad |\langle E_{n,\lambda}, E_{k,\lambda}\rangle| \le Ce^{-c2^{|n-k|}} \quad (n \ne k).$$

Consequently, uniformly in λ,

$$\left\| \frac{\sum_{n\ge1} t^n E_{n,\lambda}}{\sum_{n\ge1} t^n} \right\|_2 \le C\sqrt{1-t}, \qquad \frac{1}{2} < t < 1. \tag{5}$$

Proof. The Fourier expansion of the Poisson kernel and the root-of-unity filter give

$$\widehat{E}_{n,\lambda}(k) = r_n^{|k|} \sum_{\ell\ne0} \widehat{w}_\lambda(k + \ell M_n). \tag{6}$$

Write $k = aM_n + u$ with $|u| \le M_n/2$. Exponential decay of $\widehat{w}_\lambda$ shows that the sum in (6) is at most

$$C\big(\mathbf{1}_{a\ne0}\rho^{|u|} + \rho^{M_n-|u|}\big).$$

Moreover,

$$r_n^{M_n} = \left(\frac{1 - M_n^{-1}}{1 + M_n^{-1}}\right)^{M_n} \le e^{-2}.$$

Summing by the blocks $aM_n + [-M_n/2, M_n/2]$ therefore gives the uniform norm bound and, more precisely,

$$\|\mathbf{1}_{\{|k|<M_n/2\}} \widehat{E}_{n,\lambda}\|_{\ell^2} \le Ce^{-cM_n}, \tag{7}$$

$$\|\mathbf{1}_{\{|k|\ge LM_n\}} \widehat{E}_{n,\lambda}\|_{\ell^2} \le Ce^{-cL} \qquad (L \ge 1/2). \tag{8}$$

Suppose $n < k$. Split $E_{k,\lambda}$ at frequency $M_k/2$. Its low-frequency part is $O(e^{-cM_k})$ by (7). On the complement, the $E_{n,\lambda}$ factor is a tail with $L = M_k/(2M_n) = 2^{k-n-1}$, so (8) and Cauchy–Schwarz give

$$|\langle E_{n,\lambda}, E_{k,\lambda}\rangle| \le Ce^{-c2^{k-n}}.$$

Finally, the summability of e^{-c2^d} yields

$$\left\| \sum_{n\ge1} t^n E_{n,\lambda} \right\|_2^2 \le C \sum_{n\ge1} t^{2n} \le \frac{C}{1-t^2}.$$

Since $\sum_{n\ge1} t^n = t/(1-t)$, this is (5). ∎

3.4 Convergence of the boundary kernels

For z in a fixed compact K and $1 \le s \le 3/2$, the functions $w_{s,z} = P_z^s$ extend holomorphically to one common annulus around $\mathbb{T}$ after choosing the positive branch there. Their Fourier coefficients therefore satisfy the hypothesis of Lemma 2 uniformly in (s, z).

Let $Q_{n,s,z}$ and $C_{n,s,z}$ be the quantities in that lemma for $w_{s,z}$. From (3), positivity of Poisson kernels, and

$$\left\| \frac{1}{M_n} \sum_{j=0}^{M_n-1} P_{r_n \zeta_{n,j}} \right\|_2^2 = 1 + 2\frac{r_n^{2M_n}}{1 - r_n^{2M_n}} \le C,$$

we obtain

$$\|F_{n,s,z} - Q_{n,s,z}\|_2 \le C_K 2^{-n}. \tag{9}$$

Also $C_{n,s,z} = P_{r_n} * w_{s,z}$, so the Poisson approximate identity gives

$$\sup_{z\in K,\, 1\le s\le 3/2} \|C_{n,s,z} - w_{s,z}\|_2 \longrightarrow 0. \tag{10}$$

Write $A(t) = \sum_{n\ge1} t^n$. Combining (5), (9), (10), and the elementary Abelian fact that normalized Abel averages preserve a uniform limit, we get

$$\sup_{z\in K} \left\| \frac{\mathcal{K}_{s,z}}{A(t)} - P_z^s \right\|_2 \longrightarrow 0 \qquad (s \downarrow 1). \tag{11}$$

Taking integrals in (11) gives

$$\frac{D_{s,z}}{A(t)} - \int_{\mathbb{T}} P_z^s \, dm \longrightarrow 0 \quad \text{uniformly for } z \in K.$$

Since $\int P_z^s \, dm \to 1$ and $P_z^s \to P_z$ in L^2, uniformly on compact z-sets, (4) and (11) imply

$$\sup_{z\in K} \left\| \frac{\mathcal{K}_{s,z}}{D_{s,z}} - P_z \right\|_2 \longrightarrow 0. \tag{12}$$

Every bounded complex harmonic f has a radial boundary function $f^* \in L^\infty(\mathbb{T})$ with $f = P[f^*]$ [3, Chapter I]. Tonelli's theorem and absolute convergence give

$$\frac{g_{X_{\mathrm{dyad}}}(s, z; f)}{g_{X_{\mathrm{dyad}}}(s, z)} - f(z) = \int_{\mathbb{T}} f^*(\eta) \left(\frac{\mathcal{K}_{s,z}(\eta)}{D_{s,z}} - P_z(\eta) \right) dm(\eta).$$

The absolute value is at most $\|f\|_\infty$ times the L^1 norm of the kernel difference, which is bounded by its L^2 norm. Equation (12) proves (1) and the theorem. □

4 Verification and review priorities

- The exact $z = 0$ alias Gram matrix is

$$\langle E_n, E_k \rangle = 2\frac{(r_n r_k)^{2^{\max(n,k)}}}{1 - (r_n r_k)^{2^{\max(n,k)}}}.$$

  The included script evaluates the normalized Gram sum. The measured error divided by $\sqrt{s-1}$ approaches approximately 0.128, matching Lemma 2.

- The same script checks (3) over 2048 boundary angles for a nonreal base point. This is a numerical sanity check, not part of the proof.
- The main human-review priority is the block estimate (7)–(8) and its uniformity for P_z^s on compact z-sets. A secondary priority is whether the source's informal phrase "critical upper density" is intended in a convention stronger than the standard uniform hyperbolic ball-counting bound proved here. A literal bound by the area of every arbitrary compact Borel set cannot hold for any nonempty discrete configuration.

5 Novelty and limitations

The literature check was bounded as follows. The run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:1806.02306, arXiv:2101.09622, and the core deterministic/critical-density phrases. The full 2000–present source corpus was searched for the exact phrase "critical upper density" and the sharpened question. Web/arXiv searches used the exact question phrase together with "Patterson–Sullivan", "deterministic configuration", "simultaneous interpolation", and H^∞. They found the two Bufetov–Qiu papers but no later answer or matching dyadic construction. Thus the mathematical confidence is high subject to review, while the novelty confidence is moderate rather than certified.

The theorem answers the exact H^∞ question, and in fact all bounded harmonic functions. It does not claim simultaneous reconstruction for the larger spaces $\mathcal{H}^2(\mathbb{D};\mu)$ associated with arbitrary singular boundary measures; the $L^2(m)$ alias cancellation used here does not imply $L^2(\mu)$ cancellation.

References

[1] A. I. Bufetov and Y. Qiu, *Patterson–Sullivan measures for point processes and the reconstruction of harmonic functions*, arXiv:1806.02306v2 (2019).

[2] A. I. Bufetov and Y. Qiu, *The Patterson–Sullivan Interpolation of Pluriharmonic Functions for Determinantal Point Processes on Complex Hyperbolic Spaces*, arXiv:2101.09622v1 (2021).

[3] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.